

Exp. 5	CLOUD MIGRATION AND BACKUP SETUP USING AMAZON S3
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AIM

To perform cloud migration and backup operations by storing backup files in Amazon S3, automating backup schedules, and restoring data to verify integrity.

OBJECTIVES

- To understand cloud migration concepts.
- To understand cloud-based backup solutions.
- To create and configure Amazon S3 storage.
- To upload and restore backup data from the cloud.
- To automate backup operations.
- To verify data integrity after restoration.

THEORY

Cloud Migration is the process of moving data, applications, and workloads from local infrastructure to cloud environments.

Cloud Backup is the process of storing copies of data in cloud storage services to ensure recovery in case of hardware failure, accidental deletion, cyber-attacks, or disasters. Amazon S3 (Simple Storage Service) is a highly scalable object storage service provided by AWS. It offers durability, availability, security, and lifecycle management features suitable for backup and archival purposes. In this experiment, local files are backed up and uploaded to Amazon S3. The backup process is automated using Linux scheduling tools and data is restored to verify successful recovery.

ARCHITECTURE

Source Data



Backup Archive Creation



AWS CLI



Amazon S3 Bucket



Cloud Storage



Restore Process



Data Verification

ALGORITHM

1. Create AWS account.
2. Create S3 bucket.
3. Install AWS CLI.
4. Configure AWS credentials.
5. Create backup archive.
6. Upload backup to Amazon S3.
7. Verify uploaded files.
8. Restore files from S3.
9. Verify data integrity.
10. Automate backup using cron.

FLOWCHART

Start



Create S3 Bucket



Install AWS CLI



Configure AWS Credentials



Create Backup



Upload to S3



Verify Upload



Restore Data



Verify Integrity



Schedule Automated Backup



stop

PROCEDURE

STEP 1: LOGIN TO AWS CONSOLE

1. Login to AWS Management Console.
2. Navigate to Amazon S3.
3. Open S3 Dashboard.

STEP 2: CREATE S3 BUCKET

1. Click **Create Bucket**.
2. Enter bucket name:

cloud-backup-lab-bucket

3. Select Region.
4. Keep default settings.
5. Click **Create Bucket**.

STEP 3: INSTALL AWS CLI

Update system:

```
sudo apt update
```

Install AWS CLI:

```
sudo apt install awscli -y
```

Verify installation:

```
aws --version
```

STEP 4: CONFIGURE AWS CLI

```
aws configure
```

Provide:

- AWS Access Key ID
- AWS Secret Access Key
- Region Name
- Output Format

STEP 5: CREATE SAMPLE DATA

```
mkdir migration_data  
  
cd migration_data  
  
echo "Cloud Computing Lab Backup" > file1.txt  
echo "AWS S3 Backup Experiment" > file2.txt
```

STEP 6: CREATE BACKUP ARCHIVE

```
tar -czvf backup.tar.gz migration_data
```

STEP 7: UPLOAD BACKUP TO AMAZON S3

```
aws s3 cp backup.tar.gz s3://cloud-backup-lab-bucket/
```

STEP 8: VERIFY UPLOAD

```
aws s3 ls s3://cloud-backup-lab-bucket/
```

Expected Output:

```
backup.tar.gz
```

STEP 9: SIMULATE RESTORE OPERATION

Create restore directory:

```
mkdir restore_data
```

Download backup from S3:

```
aws s3 cp s3://cloud-backup-lab-bucket/backup.tar.gz .
```

Extract backup:

```
tar -xzf backup.tar.gz -C restore_data
```

STEP 10: VERIFY RESTORED FILES

```
ls restore_data/migration_data
```

Expected Output:

```
file1.txt  
file2.txt
```

STEP 11: AUTOMATE BACKUP PROCESS

Create backup script:

```
nano backup_script.sh
```

Script Content:

```
#!/bin/bash  
  
tar -czvf backup.tar.gz migration_data  
  
aws s3 cp backup.tar.gz s3://cloud-backup-lab-bucket/
```

Provide permission:

```
chmod +x backup_script.sh
```

Open cron editor:

```
crontab -e
```

Schedule daily backup:

```
0 10 * * * /home/user/backup_script.sh
```

COMMANDS USED

```
aws --version  
  
aws configure  
  
aws s3 ls  
  
aws s3 cp  
  
tar -czvf  
  
tar -xzvf  
  
chmod +x  
  
crontab -e
```

OBSERVATIONS

Parameter	Status
S3 Bucket Creation	Successful
AWS CLI Installation	Successful
AWS Configuration	Successful
Backup Creation	Successful
Upload to S3	Successful
Restore from S3	Successful
Data Verification	Successful
Automation Setup	Successful

EXPECTED OUTPUT

- Amazon S3 bucket created successfully.
- Backup archive uploaded to cloud storage.
- Files restored successfully from S3.
- Data integrity verified.
- Automated backup schedule configured.

RESULT

A cloud migration and backup solution was successfully implemented using Amazon S3. Backup files were uploaded to cloud storage, restored successfully, and automated backup scheduling was configured.